



BITS Pilani
Pilani Campus

Machine Learning

AIML CZG565

Unsupervised Learning

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Profs.Sugata, Chetana, Rajavadhana, Monali, Seetha, Vimal, Swarna

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Module –Unsupervised Learning

- K-means Clustering
- Mixture Models ✓
- Expectation Maximization (EM) Algorithm

$E \rightarrow \mu \rightarrow$ cluster centroid fix
calc $\rightarrow$ cluster membership
 $M \rightarrow$ mem. const
calc. μ

} $\downarrow$
min J

GMM equations

2 → π_1, π_2
 μ_1, μ_2
 Σ_1, Σ_2

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z})$$

Mixture models

K = number of mixture components (clusters),

π_j = mixture weights → ✓

$p_j(\mathbf{x})$ = family of distributions

(Gaussian, Bernoulli, etc).

$$p(\mathbf{x}) = \sum_{j=1}^K \pi_j p_j(\mathbf{x})$$

$$0 \leq \pi_j \leq 1, \quad \sum_{j=1}^K \pi_j = 1$$

Gaussian Mixture models

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$$

$\boldsymbol{\theta}$ = collection of all the parameters of the model
 (mixture weights, means, and covariance
 matrices):

$$\boldsymbol{\theta} = \left[\underbrace{\pi_1, \dots, \pi_K}_{\text{weights}}, \underbrace{\mu_1, \dots, \mu_K}_{\text{means}}, \underbrace{\Sigma_1, \dots, \Sigma_K}_{\text{covariance matrices}} \right]$$

General EM algorithm

Observed data: $X = \{x_1, \dots, x_n\}$

Unknown variables: y

In clustering: $y = 1 \dots K$ clusters

Parameters: θ

In GMM : $\theta = \pi : \{\pi_1, \dots, \pi_K\}$

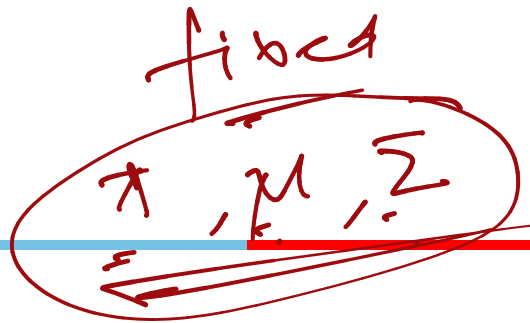
$\mu : \{\mu_1, \dots, \mu_K\}$

$\Sigma : \{\Sigma_1, \dots, \Sigma_K\}$

Goal: $\hat{\theta}_n = \arg \max_{\theta} \log P(X | \theta)$

$$\text{In GMM : } \ln p(\mathbf{X} | \pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

EM algorithm for GMM



E step:

Evaluate the responsibilities (posterior probabilities) using the current parameter

$$\begin{aligned}
 \gamma(z_k) &\equiv p(z_k = 1 | \mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1)p(\mathbf{x} | z_j = 1)} \\
 &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)} = \frac{\text{wel. prob. } k}{\text{total prob.}}
 \end{aligned}$$

Handwritten notes: x_n is written next to the first equation. γ is written next to the second equation. A red arrow points from the first equation to the second. A red box highlights the numerator of the second equation. A red arrow points from the box to the final fraction.

- For each example x_n ,
- Compute $\gamma(z_{nk})$ the probability that x_n is generated by component Z_k i.e it belongs to cluster k
- If x_n is very likely under the k_{th} Gaussian, it gets high weight



EM algorithm for GMM

M-Step :

- maximize the expectation of the complete-data log-likelihood, computed with respect to the conditional probabilities found in the Expectation step. The result of the maximization is a new parameter vector μ_{new} , Σ_{new} and π_{new}
- Keep γ (z_{nk}) fixed, and apply MLE for maximizing of $\ln p(\mathbf{X}|\pi, \mu, \Sigma)$ for μ_k , Σ_k and π_k , to get μ_{new} , Σ_{new} and π_{new}

Repeat E & M until convergence

- In practice, the algorithm is deemed to have converged when the change in the log likelihood function, or alternatively in the parameters, falls below some threshold

$$\ln p(\mathbf{X}|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

EM algorithm for GMM

To Estimate:

M-Step

Initialize π, μ, Σ and
also evaluate the log likelihood

E-Step

$$\begin{aligned}\mu_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \\ \Sigma_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T \\ \pi_k^{\text{new}} &= \frac{N_k}{N} \\ N_k &= \sum_{n=1}^N \gamma(z_{nk})\end{aligned}$$

Perform E-Step Given $\gamma(z_k)$

$$\begin{aligned}\gamma(z_k) &\equiv p(z_k = 1 | \mathbf{x}) = \\ &= \frac{p(z_k = 1) p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1) p(\mathbf{x} | z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)}\end{aligned}$$

Perform M-Step Given π, μ, Σ

$$p(z_k = 1) = \pi_k$$

Repeat Until
Convergence

N_k = the effective number of points assigned to cluster k

EM Algorithm

$K = 2$

data

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	innovate	achieve	lead
	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

initial guess

I Standardize the data if required:

II Fix the no.of.cluster expected

III Initialize the prototypes:
Mean, Covariance, Weights

IV Expectation-Step: Fix prototype & find
the membership of each point weighted
by the probability value

V. Calculate the log likelihood of the
points

VI Maximization Step: Fix the
membership(responsibility matrix) and
re-estimate the prototypes

VII Calculate the new log likelihood of
the points. Repeat E & M Step till
convergence is achieved:

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

Sample calculation for this step is shown below:

$$\begin{aligned}
 \text{Weight1} * N1 &= \\
 0.5 * P(\text{Doc1}, \text{Cluster1}) &= (\text{Doc1}, (3,3), (4,0.707)) \\
 &= 0.5 * \frac{e^{\frac{-(2-3)^2}{2*4^2}}}{4 * \sqrt{2*3.14}} * \frac{e^{\frac{-(2-3)^2}{2*0.707^2}}}{0.707 * \sqrt{2*3.14}} \\
 &= 0.01004
 \end{aligned}$$

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

For Cluster 1 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity, assume off diagonal values in covariance matrix as 0 (This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

GMM

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Similarly:

For Cluster 2 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity, assume off diagonal values in covariance matrix as 0 (This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

$$\gamma = \frac{\text{wA prob}}{\sum \text{prob.}}$$



	Doc1 (2,2)	Doc2	Doc3
$(3,3)$ $\pi_1 N_1$ doc1/ cluster1	$0.5 \times \frac{e^{-\frac{1}{2} \left(\frac{2-3}{\frac{1}{4}} \right)^2}}{\sqrt{4\sqrt{2\pi}}} \times \frac{e^{-\frac{1}{2} \left(\frac{2-3}{0.707} \right)^2}}{0.707\sqrt{2\pi}}$ $\Rightarrow 0.01004$	$0.5 \times \frac{e^{-\frac{1}{2} \left(\frac{4-3}{2} \right)^2}}{4\sqrt{2\pi}} \times \frac{e^{-\frac{1}{2} \left(\frac{4-3}{1.732} \right)^2}}{1.732\sqrt{2\pi}}$ $= 0.0005$	$\frac{e^{-\frac{1}{2} \left(\frac{5-3}{1.732} \right)^2}}{1.732\sqrt{2\pi}}$
$\pi_2 N_2$	$0.5 \times \frac{e^{-\frac{1}{2} \left(\frac{2-4}{0.5} \right)^2}}{0.5\sqrt{2\pi}} \times \frac{e^{-\frac{1}{2} \left(\frac{2-4}{0.707} \right)^2}}{0.707\sqrt{2\pi}}$ $\rightarrow = 0.00001$	0.0828	
Σ	$= 0.01005$ $\rightarrow$	0.0833	0.0063

$$\pi_1, N_1 = \pi_1 \times \mathcal{N}(\text{doc}_1)$$

$\nwarrow \quad \searrow$
 $w_1 \quad w_2$
 $(\mu_1, \Sigma_1) \quad (\mu_2, \Sigma_2)$
 $\leftarrow \mathcal{N}_1 \quad \mathcal{N}_2$

$$\pi_1, \mathcal{N}_1, \mathcal{N}_2 \quad] \text{doc}_1 \quad (3^2)$$

Assume: w_1 & w_2
are independent
 $\Rightarrow$ not correlated

$$(3, 4)$$

$$(3, 0.707)$$

$$\pi_2 N_2 = \pi_2 \cdot \int_{w_1} \int_{w_2} \text{doc}_1 / \text{chunk}_2$$

$(\mu, 0.5)$ $(\mu, 0.75)$
 (μ, Σ)

$$\text{doc}_1 \rightarrow (2, 2)$$

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	innovate	achieve	lead
	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

$$P(z_{11} = 1) = \frac{0.01004}{0.01005} =$$

For Doc2 :

$$\begin{aligned} \text{Weight1} * N1 &= 0.0005 \checkmark \\ \text{Weight2} * N2 &= 0.0828 \checkmark \\ \text{Sum} &= 0.08329 = \sim 0.0833 \checkmark \end{aligned}$$

$$P(Z_{\text{Doc2-Cluster1}}) = 0.006$$

$$P(Z_{\text{Doc2-Cluster2}}) = 0.994$$

	Doc1	Doc2	Doc3
Weight1 * N1	✓ 1	/	/
Weight2 * N2	✓ 2	/	/
Sum	✓ 3	/	/
P(Zi1)	✓ $\pi_1 N_1 / \sum$	0.006	
P(Zi2)	✓ $\pi_2 N_2 / \sum$	0.994	

Per Docs sum the previous two found values. This gives the denominator of this formula

Now find the membership of doc :

P(Zi1) = dividing Weight1*N1 by Sum

P(Zi2) = dividing Weight2*N2 by Sum

Similarly do it for all the Doc's

$$d2 \left[P(z_{21} = 1) = \frac{0.0005}{0.0833} = 0.006 \right. \\ \left. P(z_{22} = 1) = \frac{0.0828}{0.0833} = 0.994 \right]$$

GMM

V. Calculate the log likelihood of the points

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

obj = log. like

Log likelihood =
 $\ln(0.010) + \ln(0.083) + \ln(0.006)$
 = -12.16

-12.16

	Word1	Word2		innovate	achieve	lead
Doc1	2	2				
Doc2	4	5				
Doc3	7	2				
Cluster 1	(3,3)	(4,0.707)	0.5			
Cluster 2	(4,4)	(0.5,0.707)	0.5			

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Log likelihood = $\ln(\text{sum of Doc1}) + \ln(\text{sum of Doc2}) + \ln(\text{Sum of Doc3}) \dots$

This value needs to be found for every iteration and at one iteration it will approach a very large value . That will be one of the stopping convergence point of the algorithm.

EM algorithm for GMM

To Estimate:

M-Step

Initialize π, μ, Σ and
also evaluate the log likelihood

E-Step

$$\begin{aligned}\mu_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \\ \Sigma_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T \\ \pi_k^{\text{new}} &= \frac{N_k}{N} \\ N_k &= \sum_{n=1}^N \gamma(z_{nk})\end{aligned}$$

Perform E-Step Given $\gamma(z_k)$

$$\begin{aligned}\gamma(z_k) &\equiv p(z_k = 1 | \mathbf{x}) = \\ &= \frac{p(z_k = 1) p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1) p(\mathbf{x} | z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)}.\end{aligned}$$

Perform M-Step Given π, μ, Σ

$$p(z_k = 1) = \pi_k$$

Repeat Until
Convergence

N_k = the effective number of points assigned to cluster k

GMM

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

Sample calculations shown below

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

	Doc1	Doc2	Doc3
P(Zi1)	0.9999	0.006	1
P(Zi2)	0.0001	0.994	0

	Means	SDs	Weight
Cluster 1			
Cluster 2			

$N_1 =$

$$N_{k=1} = \text{Cluster 1} = 0.9999 + 0.006 + 1 = 2.0059$$

This is the value N_1 represented here

$$\text{MEAN}_{k=1} = (4.49, 2.00) = \sim (4.49, 2)$$

$$\text{Std.Dev}_{k=1} =$$

$$([0.9999(2-4.49)^2 + 0.006(4-4.49)^2 + 1(7-4.49)^2]/N_1, [0.9999(2-2)^2 + 0.006(5-2)^2 + 1(2-2)^2]/N_1)$$

$$= (6.23, 0.03)$$

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

N_1 for cluster $k=1$ is got by summing these. Similarly do it for N_2

For cluster $k=1$ is Means obtained by multiplying the data by the membership.

$$[0.9999(2) + 0.006(4) + 1(7)]/N_1, [0.9999(2) + 0.006(5) + 1(2)]/N_1$$

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

Similarly do it for cluster 2

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$$\text{Weight}_{k=\text{cluster1 (new)}} = 2.0059/3 = 0.67$$

Cluster 1



(3, 3)

μ_1, w_1	$= \frac{0.9999(2) + 0.0006(4) + 1(\cancel{7})}{2.0059} = 4.49$	
μ_1, w_2	$= \frac{0.9999(2) + 0.0006(5) + 1(2)}{2.0059} = 2$ $\textcircled{1} \mu_1^{\text{New}} = \underline{(4.49, 2)}$	
$\textcircled{2} \pi_1^{\text{New}}$	$= \frac{N_1}{N} = \frac{2.0059}{3} = \underline{0.67}$	

③ Σ

$$\mu_1 = 2.0059$$

$$\mu_1^{nw} = 4.49$$

innovate

achieve

lead

$$\Sigma_{\sigma_1, w_1}^2 = \frac{1}{2.0059} \left[0.9999 (2 - 4.49)^2 + 0.0006 (4 - 4.49)^2 + 1(2 - 4.49)^2 \right]$$

$$= 6.23$$

$$\sigma_{\sigma_1, w_2}^2 = \frac{1}{2.0059} \left[0.9999 (2 - 2)^2 + 0.0006 (5 - 2)^2 + 1(2 - 2)^2 \right]$$

$$= 0.027$$

$$\sigma_1^{nw} = (6.23, 0.027) / \sigma_1^{nw} (2.5, 0.164)$$

GMM - EM Algorithm

2nd Iteration continues



	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

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Mean, Covariance, Weights

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of the points. Repeat E & M Step till
convergence is achieved:

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Means	SDs	Weight
Cluster 1	(4.49, 2)	(6.32, 0.03)	0.67
Cluster 2	(3.99, 4.99)	(0.001, 0.001)	0.33

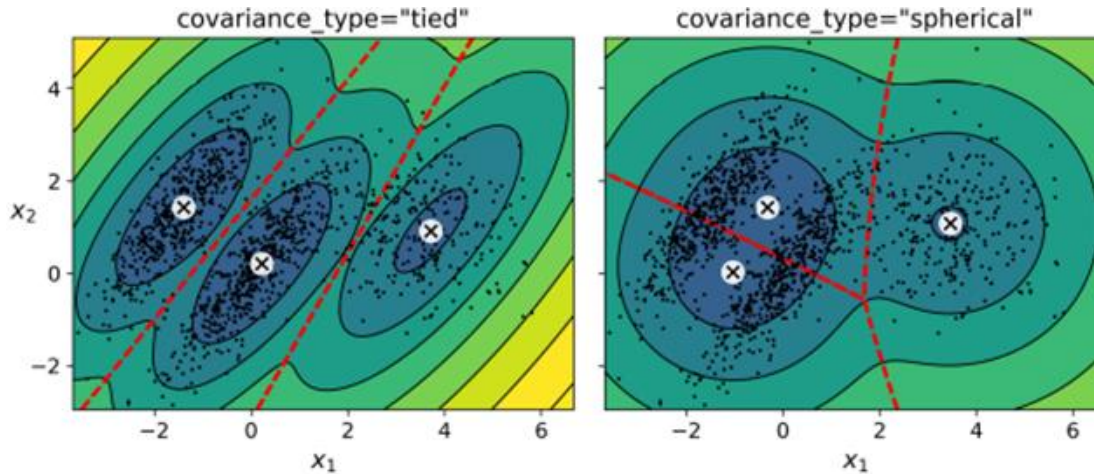
	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Gaussian Mixture Model



✓

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \leftarrow \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$



Note:

The GMM model can be relaxed by setting these parameters while calling the function

- **"spherical"**: all clusters must be spherical, but they can have different diameters (i.e., different variances).
- **"diag"**: clusters can take on any ellipsoidal shape of any size, but the ellipsoid's axes must be parallel to the coordinate axes (i.e., the covariance matrices must be diagonal).
- **"tied"**: all clusters must have the same ellipsoidal shape, size and orientation (i.e., all clusters share the same covariance matrix).

Gaussian Mixture Model

```
from sklearn.mixture import GaussianMixture
```

```
gm = GaussianMixture(n_components=3, n_init=10)  
gm.fit(X)
```

```
gm.weights_  
gm.means_  
gm.covariances_
```

```
>>> gm.predict(X)  
array([2, 2, 1, ..., 0, 0, 0])  
>>> gm.predict_proba(X)  
array([[2.32389467e-02, 6.77397850e-07, 9.76760376e-01],  
       [1.64685609e-02, 6.75361303e-04, 9.82856078e-01],
```

References

- Christopher Bishop: Pattern Recognition and Machine Learning, Springer International Edition - Chapter 9
- <https://www.youtube.com/watch?v=TG6Bh-NFhA0>
- <https://www.youtube.com/watch?v=qMTuMa86NzU>